Ryan Epperly

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EDUCATION

Virginia Tech Jan 2026 – May 2028
Bachelor of Science in Aerospace Engineering
- **Specialization**: Dynamics, Control, and Estimation | **Minor**: Mathematics

New River Community College Aug 2024 – Dec 2025
Associate of Science (A.S), Engineering | Magna Cum Laude

SKILLS & CERTIFICATIONS

Software & Programming : SolidWorks, AutoCAD, Autodesk Inventor, Ansys, MATLAB, Python, Java, GNU Octave, LaTeX, PrusaSlicer

Hardware & Testing : 3D Printing (Bambu Lab P2S, FDM/SLA), CNC Machining, Plasma, Laser & Waterjet Cutters, Glowforge, Mecmesin

Certifications : FAA UAS Safety Test, NASA Open Science 101, Ansys Topology Optimization, FAA NAFI SMS, FEMA (IS-66, IS-922.A), UCAR Satellite Remote Sensing

EXPERIENCE

Butler Parachute Systems May 2026 – Aug 2026
Engineering Intern
- Designed a line continuity tool in SolidWorks, optimizing geometry to eliminate sharp edges and prevent cord snagging.
- Executed 150+ Mecmesin/hydraulic tensile tests and performed 10 operational drop tests with a custom load cell amplifier enclosure.
- Engineered a quality control program referencing PIA documents and MIL-STD-849D, improving inspection efficiency by 20%.
- Fabricated government-regulated hardware using a Glowforge, plasma, and laser cutters; rigged 2 personnel parachutes with an FAA master rigger.

GEOINT ISAC May 2026 – Aug 2026
Space Systems Intern
- Federal ESF Project: Evaluated the 15 primary Federal Emergency Support Functions (ESFs) used under the National Response Framework to rapidly coordinate disaster respons. Analyzed how geospatial intelligence guides these deployments by mapping critical infrastructure damage and tracking Community Lifeline degradation.
- NASA Joint Project: Executed meteorological risk analysis and geospatial mapping for Project Gestalt to advance global critical infrastructure resilience.

Blackwing Space Mar 2026 – Aug 2026
Regulatory Engineering Intern
- Analyzed CubeSat systems and compiled FCC technical materials to ensure compliance and expedite satellite mission approvals.
- Verified physical component tolerances and 3D models (SolidWorks/Fusion 360) against strict aerospace licensing standards.

PROJECTS & RESEARCH

Starlink Sentinel: Autonomous LEO Collision Avoidance May 2026
- Developed a 6-DOF orbital dynamics simulator in MATLAB using an RK4 numerical integrator to model autonomous collision avoidance.
- Implemented an Extended Kalman Filter (EKF) to process noisy ground radar measurements and predict incoming debris trajectories.

Comparative Analysis of Radiation Shielding Materials Jan 2026 – May 2026
- Evaluated radiation shielding efficacy of projected Martian construction materials by developing MATLAB simulations and 3D Inventor models.
- Determined that polyethylene rapidly reduces radiation exposure to safe baseline levels at a minimal shield thickness of 7 cm.

Space Law and Policy (VASTS at NASA) Oct 2023 – Jun 2024
- Drafted a comprehensive legal and regulatory framework proposal for a crewed Mars mission in collaboration with NASA engineers.
- Outlined strict planetary protection protocols, nuclear thermal propulsion safety requirements, and radiation exposure limits.